Aamir Hussain

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Education

Ramdeobaba University Nagpur, Maharashtra
Bachelor of Technology in Computer Science – CGPA: 9.08 *2024–2028*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Machine Learning, Cloud Computing, Operating Systems, Computer Networks

Projects

Roomly – Hotel Reservation & Operations Management Platform | GitHub | Live Demo

Java, Spring Boot, Spring Security, Spring Data JPA, React, PostgreSQL, Razorpay, Google Calendar API, JWT, Maven, Docker

- Architected a full-stack **hotel reservation platform** (**Spring Boot** REST API, **React/Vite**, **PostgreSQL/Neon**) with end-to-end booking, inventory, and role-specific dashboard modules across **4** user roles.
- Implemented **JWT** authentication with **BCrypt** hashing and **role-based access control**, and integrated **Razorpay** payments with server-side order creation and cryptographic signature verification for secure access and transactions.
- Engineered a **dynamic room pricing engine** (**Strategy Pattern**) computing real-time rates from surge demand, occupancy, and urgency, and integrating the **Google Calendar API** to auto-apply holiday-based pricing multipliers.
- Eliminated **double-booking race conditions** via **pessimistic locking** with transactional availability checks; across **3 runs of 100 concurrent requests** competing for **10 available rooms**, exactly **10 reservations committed**, **90 were safely rejected**, and **zero overselling** occurred.

SwiftShip – Distributed Order Fulfillment & Real-Time Delivery Platform | GitHub | Live Demo

Java, Spring Boot, Spring Cloud, Apache Kafka, PostgreSQL, PostGIS, WebSocket, Spring AI, Redis, Docker

- Designed a **distributed quick-commerce fulfillment platform** across **8+ Spring Boot microservices**, orchestrating order placement, stock reservation, warehouse assignment, routing, tracking, and notifications via an **event-driven Kafka** pipeline.
- Hardened service-to-service communication with **Spring Cloud Gateway**, **JWT/OAuth2**, and **Resilience4j** (circuit breakers, retries, rate limiters), sustaining availability during simulated failures and preventing cascading outages.
- Implemented **PostGIS** geospatial nearest-warehouse selection via indexed distance queries, replacing application-side lat/long scanning and cutting warehouse-selection latency to **100** ms.
- Integrated **OpenRouteService** with **Spring AI/Gemini** to compute road-aware delivery routes, dynamically selecting optimal routes by real-time ETA, distance, and delivery context.
- Built real-time order tracking and multi-channel notifications using **WebSocket**, **Redis**, **Kafka**, and **Resend**, streaming updates upto **100 concurrent authenticated clients** with **4 ms p95 latency** and **0% message loss**.

SignalRoute AI – Multimodal WhatsApp Notification Router | GitHub

Python, Pandas, Pydantic, scikit-learn, Google Gemini API, Pillow

- Built a **multimodal notification-routing CLI** that classifies WhatsApp text, images, and voice notes into **notify / digest / mute** using a deterministic policy engine layered over **Gemini**-based vision and voice extraction.
- Designed a **non-overridable safety gate** that deterministically flags OTP/credential harvesting, payment-pressure scams, impersonation, and prompt-injection attempts, ensuring personalization signals can never override a clear safety risk.
- Implemented **recipient-scoped TF-IDF historical retrieval** (scikit-learn) over **400+** prior messages to surface only relevant, non-cross-user evidence for each routing decision, with post-hoc evidence validation against known message IDs.
- Achieved **1.00** action, type, and joint accuracy on the organizer's labeled evaluation set, outperforming a schema-constrained **Gemini** reasoner (**0.90** / **0.90** / **0.80**) in a head-to-head comparison; validated the pipeline with **34** unit tests and ranked **top 18%** (365/1,983) in HackerRank Orchestrate.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, HTML
Frameworks: Spring Boot, Spring Web MVC, Spring Data JPA, Spring Security, REST APIs, WebSocket, React, Resilience4j
Databases: PostgreSQL, MySQL, Oracle, Neon, Hibernate ORM
Cloud & DevOps: Docker, Kubernetes, Google Cloud, Railway, Vercel, GitHub Actions
Messaging & Observability: Apache Kafka, Redis
Tools: Git, Maven, Postman, npm

Achievements & Certifications

Ranked top 18% (365/1,983) – HackerRank Orchestrate	Leaderboard	*HackerRank — Aug 2026*
Java Spring Framework, Spring Boot, Spring AI – GenAI (55 Hours)	Credential	*Udemy — Jul 2026*
Hackerrank Java (Basic) Certification	Credential	*HackerRank — Jul 2026*